

EXPERIMENT 6

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Semester: 5
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QUESTION A:

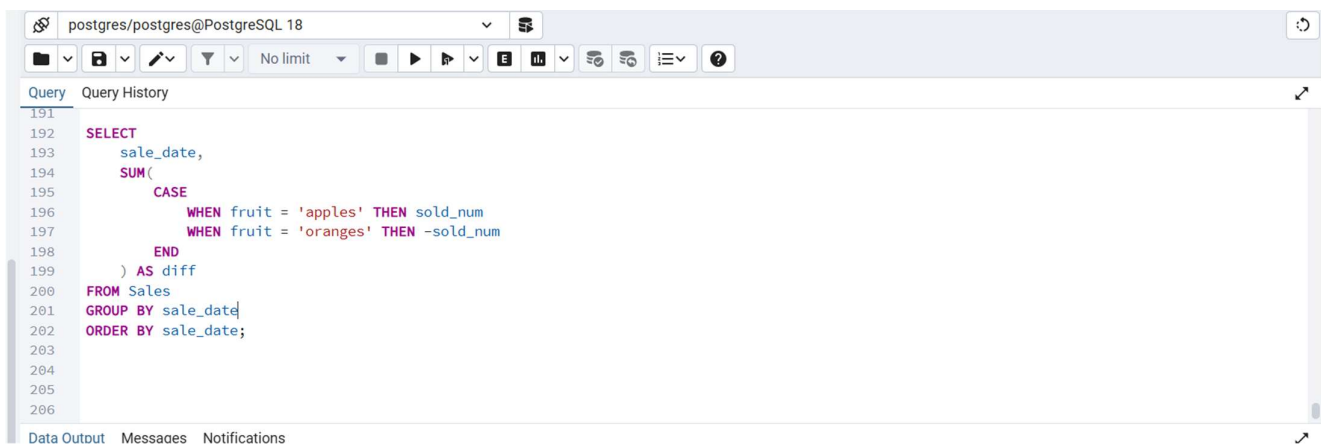
A fruit distribution company records the number of apples and oranges sold each day. For every date, exactly one record exists for apples and one record exists for oranges.

Generate a daily sales comparison report showing the difference between the number of apples sold and the number of oranges sold on the same day. The difference should be calculated as:

Difference = Apples Sold - Oranges Sold

Display the sale date and the calculated difference. The result should be sorted by sale date in ascending order.

QUERY:



```
191
192 SELECT
193     sale_date,
194     SUM(
195         CASE
196             WHEN fruit = 'apples' THEN sold_num
197             WHEN fruit = 'oranges' THEN -sold_num
198         END
199     ) AS diff
200 FROM Sales
201 GROUP BY sale_date
202 ORDER BY sale_date;
```

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RESULT:

Data Output	Messages	Notifications
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sale_date date	diff bigint	
1	2020-05-01	2
2	2020-05-02	0
3	2020-05-03	20
4	2020-05-04	-1

QUESTION B:

A company stores hierarchical employee reporting information in a table. Each record represents a node in a hierarchy where:

- id represents the current node.
- p_id represents the parent node.

Classify every node into one of the following categories:

- Root: A node that does not have any parent.
- Leaf: A node that has a parent but does not have any children.
- Inner: A node that has both a parent and at least one child.

Display the node ID along with its category. Return the result sorted by node ID.

QUERY:

```

Query Query History
229 |
230 |
231 | SELECT id,
232 |     CASE WHEN p_id IS NULL THEN 'Root'
233 |     WHEN id NOT IN (SELECT p_id FROM Tree WHERE p_id IS NOT NULL) THEN 'Leaf'
234 |     ELSE 'Inner' END as type
235 | FROM Tree
236 |
237 |
238 |
239 |
  
```

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RESULT:

[Data Output](#) Messages Notifications

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	id [PK] integer 	type text 	
1	1	Root	
2	2	Inner	
3	3	Leaf	
4	4	Leaf	
5	5	Leaf	